

ambiguities only become worse with each subsequent choice of a lift. So it is only in rather special circumstances that one can say that there are well-defined obstructions. A simple case is when $\pi_i(F) = 0$ for $i < n$, so the Moore-Postnikov factorization begins with Z_n as in the diagram at the right. In this case the composition across the bottom of the diagram gives a well-defined **primary obstruction** $\omega_n \in H^{n+1}(W, A; \pi_n F)$.

$$\begin{array}{ccccc}
 & & & \vdots & \\
 & & & \downarrow & \\
 & & & Z_{n+2} & \longrightarrow K(\pi_{n+2}F, n+3) \\
 & & \nearrow & \downarrow & \\
 A & \longrightarrow & X & \longrightarrow & Z_{n+1} \longrightarrow K(\pi_{n+1}F, n+2) \\
 \downarrow & & \searrow p & \downarrow & \\
 W & \longrightarrow & & Y = Z_n & \longrightarrow K(\pi_n F, n+1)
 \end{array}$$

Exercises

1. Show there is a map $\mathbb{R}P^\infty \rightarrow \mathbb{C}P^\infty = K(\mathbb{Z}, 2)$ which induces the trivial map on $\tilde{H}_*(-; \mathbb{Z})$ but a nontrivial map on $\tilde{H}^*(-; \mathbb{Z})$. How is this consistent with the universal coefficient theorem?
2. Show that the group structure on S^1 coming from multiplication in $\mathbb{C}$ induces a group structure on $\langle X, S^1 \rangle$ such that the bijection $\langle X, S^1 \rangle \rightarrow H^1(X; \mathbb{Z})$ of Theorem 4.57 is an isomorphism.
3. Suppose that a CW complex X contains a subcomplex S^1 such that the inclusion $S^1 \hookrightarrow X$ induces an injection $H_1(S^1; \mathbb{Z}) \rightarrow H_1(X; \mathbb{Z})$ with image a direct summand of $H_1(X; \mathbb{Z})$. Show that S^1 is a retract of X .
4. Given abelian groups G and H and CW complexes $K(G, n)$ and $K(H, n)$, show that the map $\langle K(G, n), K(H, n) \rangle \rightarrow \text{Hom}(G, H)$ sending a homotopy class $[f]$ to the induced homomorphism $f_* : \pi_n(K(G, n)) \rightarrow \pi_n(K(H, n))$ is a bijection.
5. Show that $[X, S^n] \approx H^n(X; \mathbb{Z})$ if X is an n -dimensional CW complex. [Build a $K(\mathbb{Z}, n)$ from S^n by attaching cells of dimension $\geq n+2$.]
6. For an abelian group G , Theorem 4.57 gives a map $\mu : K(G, n) \times K(G, n) \rightarrow K(G, n)$ with $\mu^*(\alpha) = \alpha \times 1 + 1 \times \alpha$ where α is a fundamental class for $K(G, n)$. Show that μ defines an H-space structure on $K(G, n)$ that is commutative and associative, up to homotopy. Show also that the H-space multiplication μ is unique up to homotopy.
7. Using an H-space multiplication μ on $K(G, n)$, define an addition in $\langle X, K(G, n) \rangle$ by $[f] + [g] = [\mu(f, g)]$ and show that under the bijection $H^n(X; G) \approx \langle X, K(G, n) \rangle$ this addition corresponds to the usual addition in cohomology.
8. Show that a map $p : E \rightarrow B$ is a fibration iff the map $\pi : E^I \rightarrow E_p$, $\pi(y) = (y(0), py)$, has a section, that is, a map $s : E_p \rightarrow E^I$ such that $ps = 1$.
9. Show that a linear projection of a 2-simplex onto one of its edges is a fibration but not a fiber bundle. [Use the preceding problem.]
10. Given a fibration $F \rightarrow E \rightarrow B$, use the homotopy lifting property to define an action of $\pi_1(E)$ on $\pi_n(F)$, a homomorphism $\pi_1(E) \rightarrow \text{Aut}(\pi_n(F))$, such that the composition $\pi_1(F) \rightarrow \pi_1(E) \rightarrow \text{Aut}(\pi_n(F))$ is the usual action of $\pi_1(F)$ on $\pi_n(F)$. Deduce that if $\pi_1(E) = 0$, then the action of $\pi_1(F)$ on $\pi_n(F)$ is trivial.

11. For a space B , let $\mathcal{F}(B)$ be the set of fiber homotopy equivalence classes of fibrations $E \rightarrow B$. Show that a map $f: B_1 \rightarrow B_2$ induces $f^*: \mathcal{F}(B_2) \rightarrow \mathcal{F}(B_1)$ depending only on the homotopy class of f , with f^* a bijection if f is a homotopy equivalence.
12. Show that for homotopic maps $f, g: A \rightarrow B$ the fibrations $E_f \rightarrow B$ and $E_g \rightarrow B$ are fiber homotopy equivalent.
13. Given a map $f: A \rightarrow B$ and a homotopy equivalence $g: C \rightarrow A$, show that the fibrations $E_f \rightarrow B$ and $E_{fg} \rightarrow B$ are fiber homotopy equivalent. [One approach is to use Corollary 0.21 to reduce to the case of deformation retractions.]
14. For a space B , let $\mathcal{M}(B)$ denote the set of equivalence classes of maps $f: A \rightarrow B$ where $f_1: A_1 \rightarrow B$ is equivalent to $f_2: A_2 \rightarrow B$ if there exists a homotopy equivalence $g: A_1 \rightarrow A_2$ such that $f_1 \simeq f_2 g$. Show the natural map $\mathcal{F}(B) \rightarrow \mathcal{M}(B)$ is a bijection. [See Exercises 11 and 13.]
15. If the fibration $p: E \rightarrow B$ is a homotopy equivalence, show that p is a fiber homotopy equivalence of E with the trivial fibration $\mathbb{1}: B \rightarrow B$.
16. Show that a map $f: X \rightarrow Y$ of connected CW complexes is a homotopy equivalence if it induces an isomorphism on π_1 and its homotopy fiber F_f has $\tilde{H}_*(F_f; \mathbb{Z}) = 0$.
17. Show that ΩX is an H-space with multiplication the composition of loops.
18. Show that a fibration sequence $\cdots \rightarrow \Omega B \rightarrow F \rightarrow E \rightarrow B$ induces a long exact sequence $\cdots \rightarrow \langle X, \Omega B \rangle \rightarrow \langle X, F \rangle \rightarrow \langle X, E \rangle \rightarrow \langle X, B \rangle$, with groups and group homomorphisms except for the last three terms, abelian groups except for the last six terms.
19. Given a fibration $F \rightarrow E \xrightarrow{p} B$, define a natural action of ΩB on the homotopy fiber F_p and use this to show that exactness at $\langle X, F \rangle$ in the long exact sequence in the preceding problem can be improved to the statement that two elements of $\langle X, F \rangle$ have the same image in $\langle X, E \rangle$ iff they are in the same orbit of the induced action of $\langle X, \Omega B \rangle$ on $\langle X, F \rangle$.
20. Show that by applying the loop space functor to a Postnikov tower for X one obtains a Postnikov tower of principal fibrations for ΩX .
21. Show that in the Postnikov tower of an H-space, all the spaces are H-spaces and the maps are H-maps, commuting with the multiplication, up to homotopy.
22. Show that a principal fibration $\Omega C \rightarrow E \xrightarrow{p} B$ is fiber homotopy equivalent to the product $\Omega C \times B$ iff it has a section, a map $s: B \rightarrow E$ with $ps = \mathbb{1}$.
23. Prove the following uniqueness result for the Quillen plus construction: Given a connected CW complex X , if there is an abelian CW complex Y and a map $X \rightarrow Y$ inducing an isomorphism $H_*(X; \mathbb{Z}) \approx H_*(Y; \mathbb{Z})$, then such a Y is unique up to homotopy equivalence. [Use Corollary 4.73 with W the mapping cylinder of $X \rightarrow Y$.]
24. In the situation of the relative lifting problem, suppose one has two different lifts $W \rightarrow X$ that agree on the subspace $A \subset W$. Show that the obstructions to finding a homotopy rel A between these two lifts lie in the groups $H^n(W, A; \pi_n F)$.

Additional Topics

4.A Basepoints and Homotopy

In the first part of this section we will use the action of π_1 on π_n to describe the difference between $\pi_n(X, x_0)$ and the set of homotopy classes of maps $S^n \rightarrow X$ without conditions on basepoints. More generally, we will compare the set $\langle Z, X \rangle$ of basepoint-preserving homotopy classes of maps $(Z, z_0) \rightarrow (X, x_0)$ with the set $[Z, X]$ of unrestricted homotopy classes of maps $Z \rightarrow X$, for Z any CW complex with basepoint z_0 a 0-cell. Then the section concludes with an extended example exhibiting some rather subtle nonfinite generation phenomena in homotopy and homology groups.

We begin by constructing an action of $\pi_1(X, x_0)$ on $\langle Z, X \rangle$ when Z is a CW complex with basepoint 0-cell z_0 . Given a loop γ in X based at x_0 and a map $f_0: (Z, z_0) \rightarrow (X, x_0)$, then by the homotopy extension property there is a homotopy $f_s: Z \rightarrow X$ of f_0 such that $f_s(z_0)$ is the loop γ . We might try to define an action of $\pi_1(X, x_0)$ on $\langle Z, X \rangle$ by $[\gamma][f_0] = [f_1]$, but this definition encounters a small problem when we compose loops. For if η is another loop at x_0 , then by applying the homotopy extension property a second time we get a homotopy of f_1 restricting to η on x_0 , and the two homotopies together give the relation $([\gamma][\eta])[f_0] = [\eta]([\gamma][f_0])$, in view of our convention that the product $\gamma\eta$ means first γ , then η . This is not quite the relation we want, but the problem is easily corrected by letting the action be an action on the right rather than on the left. Thus we set $[f_0][\gamma] = [f_1]$, and then $[f_0]([\gamma][\eta]) = ([f_0][\gamma])[\eta]$.

Let us check that this right action is well-defined. Suppose we start with maps $f_0, g_0: (Z, z_0) \rightarrow (X, x_0)$ representing the same class in $\langle Z, X \rangle$, together with homotopies f_s and g_s of f_0 and g_0 such that $f_s(z_0)$ and $g_s(z_0)$ are homotopic loops. These various homotopies define a map $H: Z \times I \times \partial I \cup Z \times \{0\} \times I \cup \{z_0\} \times I \times I \rightarrow X$ which is f_s on $Z \times I \times \{0\}$, g_s on $Z \times I \times \{1\}$, the basepoint-preserving homotopy from f_0 to g_0 on $Z \times \{0\} \times I$, and the homotopy from $f_s(z_0)$ to $g_s(z_0)$ on $\{z_0\} \times I \times I$. We would like to extend H over $Z \times I \times I$. The pair $(I \times I, I \times \partial I \cup \{0\} \times I)$ is homeomorphic to $(I \times I, I \times \{0\})$, and via this homeomorphism we can view H as a map $Z \times I \times \{0\} \cup \{z_0\} \times I \times I \rightarrow X$, that is, a map $Z \times I \rightarrow X$ with a homotopy on the subcomplex $\{z_0\} \times I$. This means the homotopy extension property can be applied to produce an extension of the original H to $Z \times I \times I$. Restricting this extended H to $Z \times \{1\} \times I$ gives a basepoint-preserving homotopy $f_1 \simeq g_1$, which shows that $[f_0][\gamma]$ is well-defined.

Note that in this argument we did not have to assume the homotopies f_s and g_s were constructed by applying the homotopy extension property. Thus we have proved